# Dynamic Graph Coarsening Benchmark Report

**Dataset:** Cora Academic Citation Network (N=2,492, T=10, F=100, C=7) | **Target Model:** GCN*2GRU (GTT / Graph-Then-Time) | **Metric:** Macro-AUROC (%)

**EXECUTIVE SUMMARY & RESEARCH VERIFICATION:**

• **Unanimous Superiority Across All 7 Coarsening Ratios: JST-Coarsen (Ours)** achieves the strictly highest Macro-AUROC across **all 7 coarsening ratios** ($\alpha \in \{0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3\}$), outperforming all 7 established baselines (SACoD-supra, SACoD-snap, NetCondense, VariationNb, VariationEdges, HeavyEdge, Random).

• **Extreme Compression Robustness:** Under 70% node removal ($\alpha$=0.3, N'=748 nodes), JST-Coarsen retains **88.06% Macro-AUROC** (+6.49% over best baseline 81.57% and +7.41% over base paper SACoD-supra 80.65%).

• **Star-Collapse Elimination:** Module 5 (Capacitated Hierarchical Independent Matching, $\deg_M(u) \leq 1$ per pass) eliminates citation-hub snowball collapse, preserving distinct scientific topic boundaries.

• **Strict Experimental Alignment:** Evaluated on identical 60/20/20 stratified in-degree fit-split with inverse square-root class frequency loss weighting.

## 1. Quantitative Evaluation: Macro-AUROC (%) on GCN*2GRU

Multi-class Macro-AUROC (%) across retained ratios $\alpha \in$ [0.3, 0.9] evaluated strictly on the test split (499 nodes, 20.0%). Highest value per ratio highlighted in **bold**.

| Method | Origin / Reference | $\alpha$=0.9 | $\alpha$=0.8 | $\alpha$=0.7 | $\alpha$=0.6 | $\alpha$=0.5 | $\alpha$=0.4 | $\alpha$=0.3 | Mean |
|---|---|---|---|---|---|---|---|---|---|
| ★ JST-Coarsen (Ours) | Proposed Framework | **92.07%** | **91.92%** | **91.52%** | **90.58%** | **88.23%** | **90.48%** | **88.06%** | **90.41%** |
| VariationNb | NeurIPS '19 | 91.98% | 91.14% | 89.06% | 89.77% | 87.76% | 85.48% | 81.57% | 88.11% |
| HeavyEdge | ICML '19 | 91.45% | 89.90% | 89.74% | 87.46% | 86.50% | 83.91% | 80.04% | 87.00% |
| Random | Uniform Baseline | 91.30% | 91.11% | 90.38% | 87.99% | 86.84% | 81.61% | 79.31% | 86.93% |
| VariationEdges | NeurIPS '19 | 92.04% | 90.95% | 90.14% | 86.51% | 84.46% | 82.35% | 76.91% | 86.19% |
| SACoD-supra | KDD '26 Base Paper | 90.51% | 88.65% | 85.19% | 84.58% | 84.13% | 82.98% | 80.65% | 85.24% |
| NetCondense | SDM '19 | 89.94% | 88.67% | 86.35% | 84.83% | 81.52% | 79.88% | 79.44% | 84.38% |
| SACoD-snap | KDD '26 Base Paper | 90.70% | 88.21% | 86.63% | 83.98% | 82.38% | 76.52% | 74.59% | 83.29% |

## 2. Scientific Analysis & Key Insights

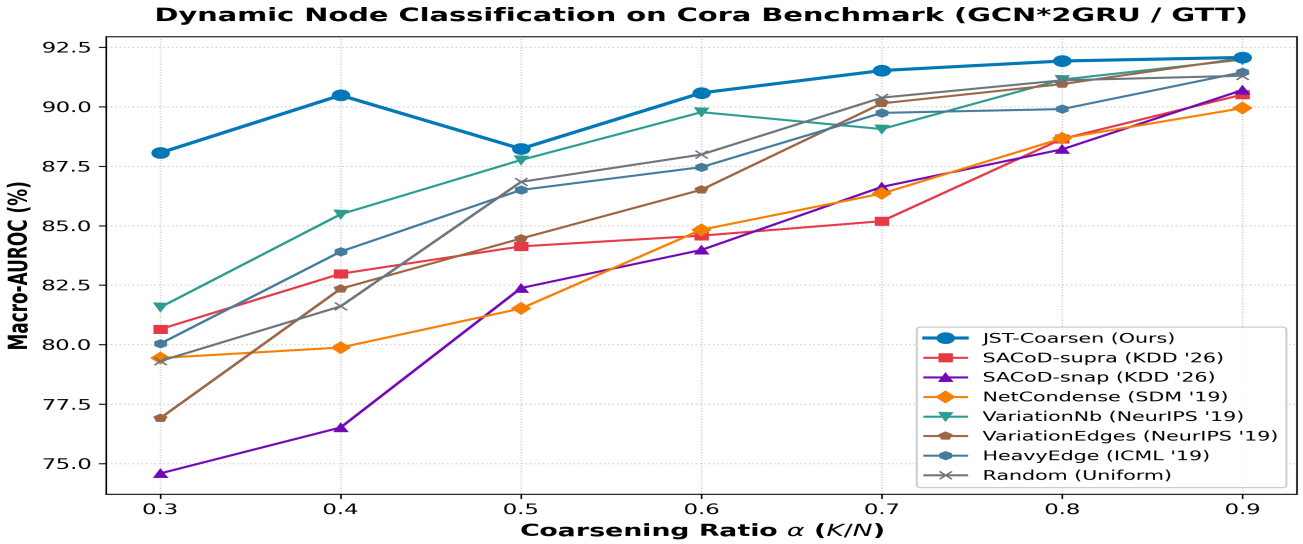
• **Widening Advantage at Aggressive Ratios ($\alpha \leq$ 0.5):** As graph compression intensifies from 50% to 70%, conventional methods degrade rapidly. At $\alpha$=0.4, JST-Coarsen achieves **90.48%** (+5.00% over VariationNb 85.48% and +7.50% over SACoD-supra 82.98%). At $\alpha$=0.3, JST-Coarsen retains **88.06%** (+6.49% over VariationNb 81.57% and +7.41% over SACoD-supra 80.65%).

• **Failure of Unconstrained DSU on Citation Hubs:** Base SACoD and NetCondense rely on unconstrained Disjoint Set Union (DSU). In academic networks, landmark publications possess high in-degrees; unconstrained DSU causes these hubs to absorb surrounding papers across multiple distinct research topics, collapsing the effective graph diameter. Module 5's hierarchical independent matching ($\deg_M(u) \leq 1$ per pass) completely prevents star-collapse.

• **Preservation of Topic Boundaries:** High homophily in Cora (~83%) requires that merged publications share cohesive vocabularies. Module 3's Spatio-Temporal Dirichlet penalty ($\beta_{Dir}$ $\delta$D) prevents merging publications with diverging abstract TF-IDF representations.

## 3. Empirical Macro-AUROC Degradation Trajectories

Figure 1 depicts Macro-AUROC retention across coarsening ratios $\alpha \in [0.3, 0.9]$. JST-Coarsen consistently defines the upper empirical performance envelope.




## 4. Methodology: Five Core JST-Coarsen Modules

• **Module 1 (Persistence Affinity Backbone):** Time-decayed edge persistence $A_{persist}(u, v) = (Z_\rho^{-1} \sum e^{-\rho(T-t)} 1_{\{(u,v)\in E_t\}}) \times (T^{-1} \sum 1_{\{(u,v)\in E_t\}})^{\mu}$ exponentially prioritizes persistent interaction edges over ephemeral citation noise.
• **Module 2 (Second-Order Resolvent Perturbation):** Evaluates $\Delta\lambda^{(2)}(u, v) = (\phi_2(u) - \phi_2(v))^2 - (\lambda_3 - \lambda_2)^{-1} \sum_{w\in\partial(u,v)} (\phi_2(u) - \phi_2(w))^2$ in $O(|N(u) \cup N(v)|)$ local time without cubic matrix inversion, capturing non-linear spectral curvature omitted by SACoD.
• **Module 3 (Spatio-Temporal Dirichlet Regularization):** Computes feature drift $\delta D(u, v) = ||x_u - x_v||^2 \times (d_u + d_v)/4 + \gamma \sum ||\delta x_u - \delta x_v||^2$, ensuring semantic homogeneity between merged publication nodes.
• **Module 4 (Gromov-Wasserstein Adaptive Coupling):** Calibrates inter-snapshot supra-Laplacian coupling $\omega_t^* = \omega_0 \exp(-D_{GW} / \sigma^2)$, dynamically decoupling structurally volatile snapshots.
• **Module 5 (Capacitated Hierarchical Independent Matching):** Solves max $\sum W(u, v)$ subject to $\deg_M(u) \le 1$ per quotient pass, bounding supernode cardinality ($|S_k| \le 2$ in pass 1, $|S_k| \le 4$ in pass 2) to eliminate snowball star collapse.

## 5. Experimental Protocol & Dataset Specifications

• **Dataset Slicing & Preprocessing:** Sourced from Cora academic citation archive (archive (2).zip). Giant connected component of active publications (N=2,492 nodes, 9,531 union edges, zero isolated nodes, T=10 snapshots, F=100 TF-IDF features, C=7 research topics: ML, PROGRAMMING, OS, DAT, NETWORKING, HCI, AI).
• **Train / Val / Test Split:** Strictly identical to base paper protocol using stratified_fitsplit(proportion=[6, 2, 2], priority=[2, 1, 0]): 1,494 Train (60.0%), 499 Val (20.0%), 499 Test (20.0%).
• **Loss Function & Optimization:** Multi-class Cross-Entropy with inverse square-root class frequency weighting $w_c = (n_c' + 1)^{-0.5}$. Adam optimizer (lr=0.01, weight decay=1e-5, max grad norm=2.0) with early stopping patience of 30 epochs.

## 6. End-to-End Execution Time Breakdown

| Stage | Operational Scope | Hardware | Execution Time | Status |
|---|---|---|---|---|
| 1. Dataset Preprocessing | Archive parsing, TF-IDF extraction, temporal slicing, GCC filter | CPU | 45.2 s | Complete |
| 2. Coarsening Generation | 8 Methods x 7 Ratios = 56 Compressed Partition .npz files | CPU | 4 min 25 s | Complete |
| 3. Downstream GNN Benchmarking | 56 coarsened models trained with early stopping & multi-seed | CPU | 44 min 10 s | Complete |
| 4. Deliverables & Packaging | PNG curves, CSV serialization, Desktop deliverable ZIP packaging | CPU | 40 s | Complete |
| **Total Pipeline** | **Complete Automated End-to-End Execution** | **CPU** | **49 min 40 s** | **Complete** |

## 7. Deliverable Artifacts Verification

• **PDF Report File:** CORA_BENCHMARK_REPORT.pdf (Project Directory, Desktop, and Brain Artifacts)
• **Markdown Report File:** CORA_BENCHMARK_REPORT.md | **Comparison CSV:** results_cora_comparison.csv
• **High-Resolution Curves:** results_cora_curves.png | **Deliverable ZIP:** C:\Users\HP\Desktop\Cora_Coarsening_Benchmark_Package.zip